

Product	Benefits	Technology	ISO viscosity grades	Specifications and approvals (Full details of approvals for all products can be obtained from your Shell representative; approvals and claims will vary by viscosity grade.)
Shell Corena S4 R	■ Extra long life and improved efficiency ■ Severe applications ■ Rotary air compressors	Synthetic, advanced ashless	32, 46, 68	Approved by: Shell Corena S4 R 68 is approved by ABB for use in VTR turbochargers with a maximum oil-drain interval of 5,000 hours (HZTL 90617, list 3a) Industry standards: ISO 6743-3A DAJ
Shell Corena S4 P	■ Extra long life and improved efficiency ■ Severe applications ■ Reciprocating air compressors	Synthetic, advanced additive system	68, 100	Industry standards: ISO 6743-3: 2003 DAB (severe duty); DIN 51506 VDL/DP 6521-L-DAB (medium duty); EN 12021
Shell Corena S3 R	■ Long life ■ Improved efficiency ■ Rotary air compressors	Conventional, advanced additive system	32, 46, 68	Industry standards: ISO 6743-3A DAH, DAJ
Shell Corena S2 R	■ Reliable protection ■ Standard life applications ■ Rotary air compressors	Conventional, detergent	46, 68	Industry standards: ISO 6743-3A DAH
Shell Corena S2 P	■ Reliable protection ■ Standard life applications ■ Reciprocating air compressors	Conventional	68, 100, 150	Industry standards: ISO 6743-3A DAA (normal duty)
Speciality compressor oils				
Shell Refrigeration Oils	A range of specialised oils suitable for use in carbon dioxide, ammonia and hydrofluorocarbon refrigeration systems		Contact your Shell representative for details	
Shell Gas Compressor Oils	A range of specialised oils suitable for use in hydrocarbon and chemical gas compression applications		Contact your Shell representative for details	

FULL PRODUCT AND SERVICE PORTFOLIO

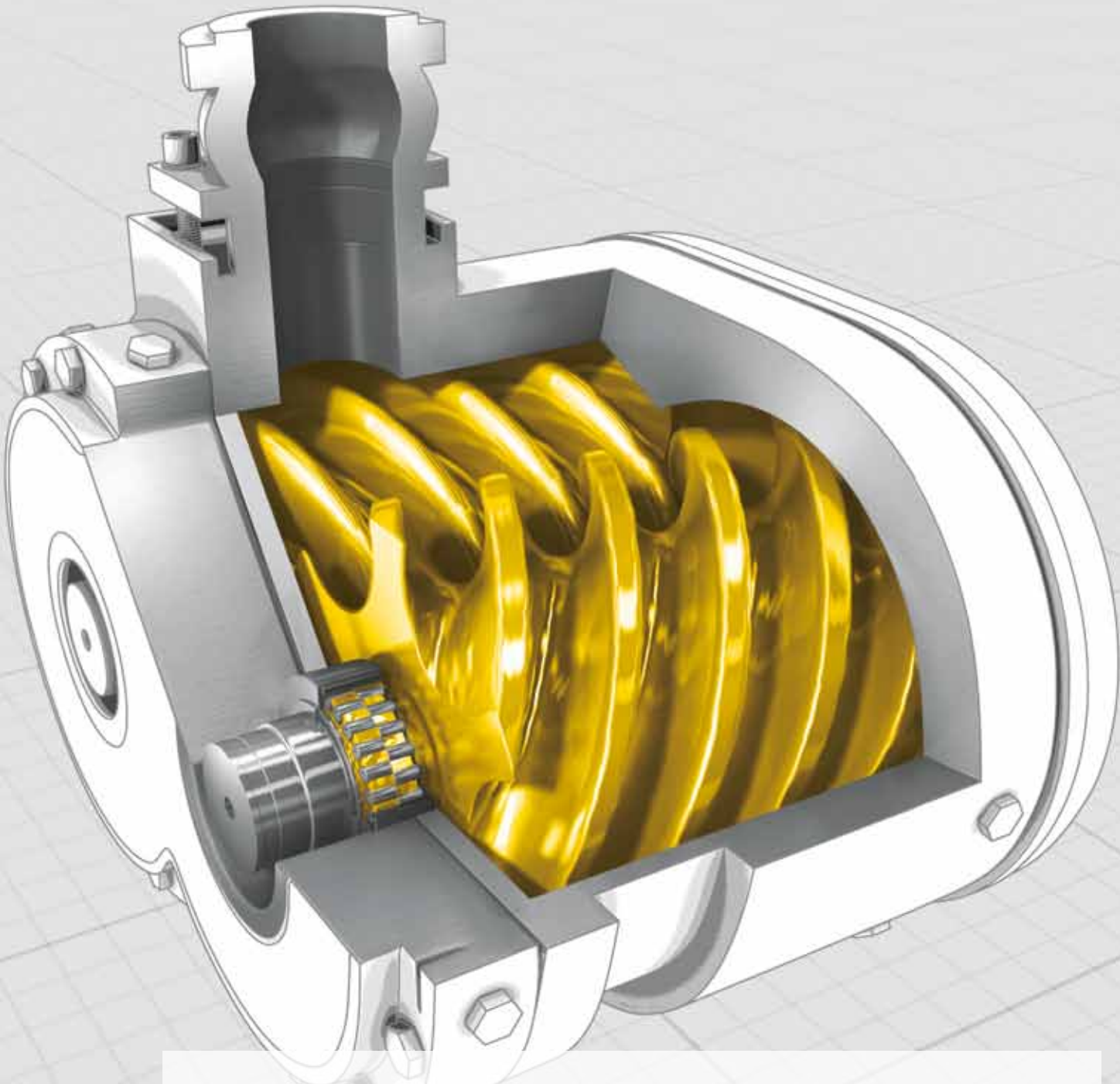
Shell Lubricants is named the market leader in lubricants [Kline & Company, 2009] and has a 60-year history of innovation. We are constantly investing to develop better lubrication solutions, as demonstrated by

- Shell Omala S4 GX synthetic gear oil – for long life in demanding environments
- Shell Tellus S4 ME synthetic hydraulic oil – for long life and energy saving.

In addition, Shell provides the excellent Shell LubeAnalyst oil condition monitoring service, which is designed to help improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.

Shell Lubricants refers to the various Shell companies engaged in the lubricants business.



DESIGNED TO ENHANCE
PRODUCTIVITY
JUST LIKE OUR COMPRESSOR
OILS – SHELL CORENA.

DESIGNED TO MEET CHALLENGES



For more information, please contact

EVERY PART OF YOUR MACHINE OR PROCESS HAS BEEN METICULOUSLY ENGINEERED, SO YOU WANT TO BE SURE THAT YOU CHOOSE A LUBRICANT THAT HAS BEEN DESIGNED TO ENSURE THAT YOUR EQUIPMENT IS WELL PROTECTED AND WORKS EFFICIENTLY.

The Shell Corena range of air compressor oils has been developed to enable equipment operators to select the oil designed to deliver optimum value to their operations through enhanced wear protection, long oil life and high system efficiency.

WEAR PROTECTION

Excess wear can seriously compromise an air compressor’s life and efficiency. The Shell Corena range includes dedicated rotary compressor oils designed to control vane and screw wear, and oils for reciprocating compressors specifically to help control cylinder, piston-ring and valve wear.

Whatever your technical needs and operational conditions, there is a Shell oil that is designed to help you get the most from your equipment investment. The Shell Corena range includes the latest synthetic oils for enhanced wear protection in high-temperature conditions and for extended operations.

OIL LIFE

The longer an oil’s life, the less fluid maintenance your equipment requires, so it can continue to operate with less interruption. The Shell Corena range enables you to best match oil life to your operational needs.

From mainstream oils for standard life applications to advanced synthetic oils that can last for up to three years, the Shell Corena range enables you to choose the right oil for your needs.

SYSTEM EFFICIENCY

Pressure loss in worn compressors can reduce operational efficiency and increase costs. Shell Corena oils are designed to promote system efficiency by suppressing foaming, releasing trapped air and rapidly separating out condensed water. The oils are also developed to help to reduce wear, which helps to keep your compressors working as their designers intended.

A RANGE OF AIR COMPRESSOR OILS DESIGNED TO MEET YOUR NEEDS

To meet the challenges of a wide range of air compressors and applications, Shell has designed a portfolio of oils that enables you to choose a product to match your needs.

PRODUCT NAME SUFFIX KEY

- P = High pressure/reciprocating (piston) air compressors
- R = Rotary vane and screw air compressors

APPLICATION ICON KEY

- High temperature
- Mobile compressors
- Rotary vane and screw air compressors
- Factory/machine applications
- Reciprocating (piston) air compressors

THE SHELL CORENA “R” RANGE

Rotary vane and screw air compressor oils for use in stationary and mobile air compressors

THE SHELL CORENA “P” RANGE

Reciprocating (piston) air compressor oils for safe and reliable operation in demanding high-pressure applications



Shell Corena S4 R

SYNTHETIC

- Extra long life
- Improved efficiency
- Severe applications

Icons: Factory, Air, High temperature, Reciprocating (piston) air compressors

Better protection, longer oil life and better system efficiency

Shell Corena S3 R

- Long life
- High efficiency

Icons: Factory, Air, Reciprocating (piston) air compressors

Better protection, longer oil life and better system efficiency

Shell Corena S2 R

- Reliable protection
- Standard life applications

Icons: Factory, Air, Reciprocating (piston) air compressors

Shell Corena S4 P

SYNTHETIC

- Extra long life
- Improved efficiency
- Severe applications

Icons: Factory, Air, High temperature, Reciprocating (piston) air compressors

Better protection and longer oil life

Shell Corena S2 P

- Reliable protection
- Standard life applications

Icons: Factory, Air, Reciprocating (piston) air compressors

PERFORMANCE YOU CAN COUNT ON

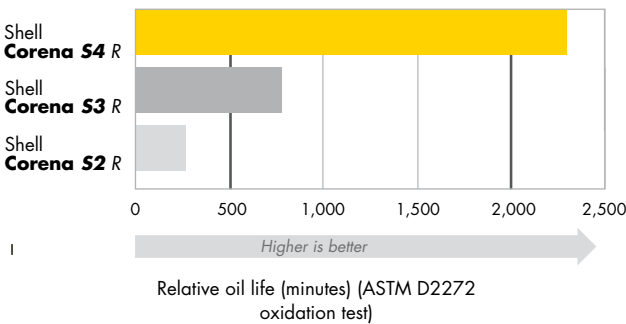
Shell’s air compressor oils are developed in close co-operation with customers and equipment makers. The Shell Corena range is well known to equipment makers, and its performance has been demonstrated in real-life applications to help reduce the possibility of your equipment letting you down.

REAL-WORLD VALUE DELIVERY

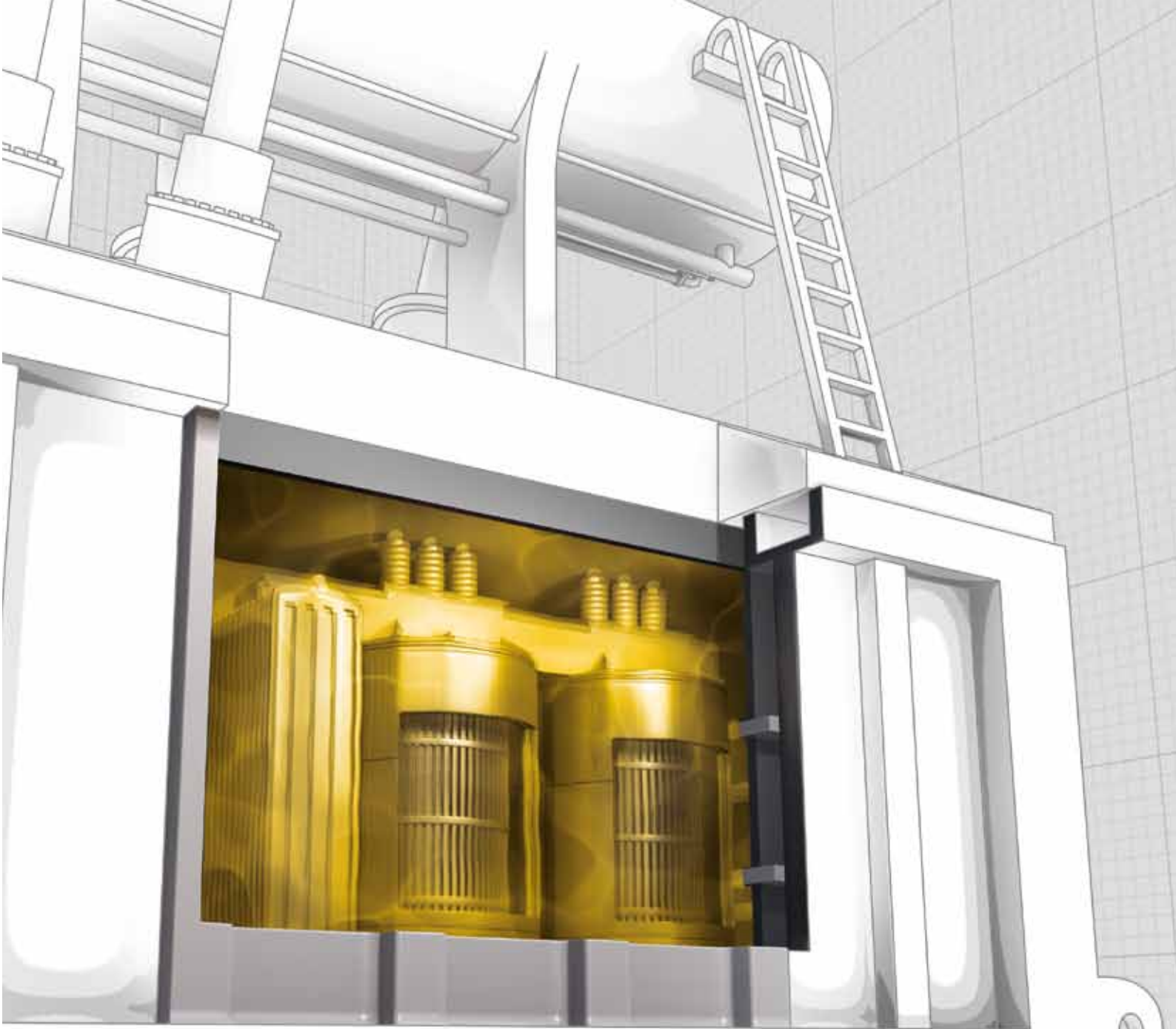
Users of Shell Corena oils across a wide range of industries are benefiting from proven lubrication performance that adds value to their operations. For instance, by switching to Shell Corena S4 R, some customers are enhancing production and cutting maintenance costs by

- increasing oil-drain intervals by 100% compared with conventional oils
- extending equipment service life: one cement company is running its compressors for three times longer between overhauls
- reducing the high operating temperatures caused by thickening of the original oil, which prevented the oil valves from working efficiently.

Shell Corena S4 R – For the longest oil life in the Shell range



Product	Category descriptor	Benefits	Technology	Specifications and approvals
Shell Diala S3 ZX-IG	Premium inhibited electrical insulating oil	■ Extra gas-absorbing performance ■ Meets IEC 60296	Inhibited	Meets the requirements of or is approved by most leading equipment manufacturers and utility companies Industry standards: IEC 60296 (2003), Table 2 Transformer Oil (I) (Inhibited oil); Baader oxidation test (obsolete) DIN 57370-1 (1978)
Shell Diala S2 ZUH	Uninhibited electrical insulating oil	■ Reliable performance ■ Meets IEC 60296	Uninhibited	Meets the requirements of or is approved by most leading equipment manufacturers and utility companies Industry standards: IEC 60296 (2003), Table 2 Transformer Oil (U) (Uninhibited oil)



**COMPREHENSIVE PRODUCT
AND SERVICE PROVISION**

Shell Lubricants is named the number one lubricants supplier [Kline & Company, 2009] and has a 60-year history of innovation. We are constantly investing to develop better lubrication solutions, as demonstrated by

- Shell Turbo GT – a fully synthetic top-tier industrial gas turbine oil
- Shell Mysella XL – for the most-demanding, natural-gas-powered generator sets.

In addition, Shell provides the excellent Shell LubeAnalyst oil condition monitoring service, which is designed to help you improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.

Shell Lubricants refers to the various Shell companies engaged in the lubricants business.



**DESIGNED TO KEEP
ON DELIVERING.
JUST LIKE OUR
ELECTRICAL INSULATING
OILS – SHELL DIALA.**

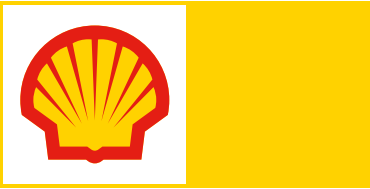
DESIGNED TO MEET CHALLENGES



For more information, please contact



[shell.com/lubricants](https://www.shell.com/lubricants)



EVERY PART OF YOUR MACHINE OR PROCESS HAS BEEN METICULOUSLY ENGINEERED, SO YOU WANT TO BE SURE THAT YOU CHOOSE A LUBRICANT THAT HAS BEEN DESIGNED TO ENSURE THAT YOUR EQUIPMENT IS WELL PROTECTED AND WORKS EFFICIENTLY.

The Shell Diala range of electrical oils has been developed to enable equipment manufacturers and power companies to select the oil designed to deliver optimum value to their operations through enhanced equipment protection, long oil life and high system efficiency.

EQUIPMENT PROTECTION

An electrical insulating oil lies at the heart of your transformer. The Shell Diala range of electrical insulating oils offers a number of options, including the latest fully inhibited, ultra-low-sulphur technology that meets all the current IEC and ASTM standards for copper corrosion, and is designed to ensure the best possible protection for your high-value assets. Shell also offers special gas-absorbing insulating oils for use in high-electrical-stress applications such as those found in high-voltage, direct-current (HVDC) transformers.

OIL LIFE

A long and predictable oil life is critical to transformer performance. The Shell Diala range of oils includes inhibited and uninhibited electrical insulating oils to give you a choice of performance profiles to best match your requirements. Shell’s latest products meet the IEC 60296 high level of oxidation performance and exceed the 500-hour oxidation test limit.

SYSTEM EFFICIENCY

To help your transformer perform to its design standards, an electrical insulating oil must maintain its characteristics over a wide range of operating temperatures. All the Shell Diala electrical oils are designed to offer excellent heat transfer and dielectric properties, even at the lowest start-up operating temperatures. In addition, Shell Diala oils are formulated to offer excellent resistance to the effects of ageing, sludge formation and deposit build-up.

A RANGE OF ELECTRICAL OILS DESIGNED TO MEET YOUR NEEDS

To meet the challenges of a wide range of equipment designs and applications, Shell has designed a portfolio of fluids that enables you to choose a product to best match your technical and operational needs.

PRODUCT-NAME SUFFIX KEY

- G = Gas absorbing
- I = Conforms to IEC specifications
- U = Uninhibited
- X = Fully inhibited
- Z = Insulating oil

APPLICATION ICON KEY

-  Transformer applications
-  Power station applications

INCREASINGLY EFFICIENT PROTECTION

ADVANCED
TIER 4

PREMIUM
TIER 3

MAIN LINE
TIER 2

THE SHELL DIALA RANGE FOR STANDARD APPLICATIONS

THE SHELL DIALA RANGE FOR SPECIAL APPLICATIONS

Shell Diala S2 ZU-I

- Reliable performance
- Meets IEC 60296



Shell Diala S3 ZX-IG

- Extra gas-absorbing performance
- Meets IEC 60296



CORROSIVENESS TO COPPER

All Shell Diala products meet the industry standards for copper corrosion: IEC 62535, ASTM D1275B and DIN 51353.

In addition, Shell is committed to minimising insulating-oil-related copper corrosion in transformers. The organisation works with equipment manufacturers, utility companies and specifying technical committees to understand the causes of corrosion and to reduce or even eliminate sulphur levels in its electrical insulating oils.

SPECIALIST APPLICATIONS

Shell offers several products designed for specialist applications. These include

- Shell Naturelle Transformer Oil S4 I. This is a synthetic-ester-based product that offers excellent low-flammability characteristics. It has a flashpoint of 275°C, is fully biodegradable and meets European Union Ecolabel requirements
- Shell Diala Switchgear Oil, which is designed for switchgear applications and is highly suitable for applications likely to operate at very low temperatures without additional heating.

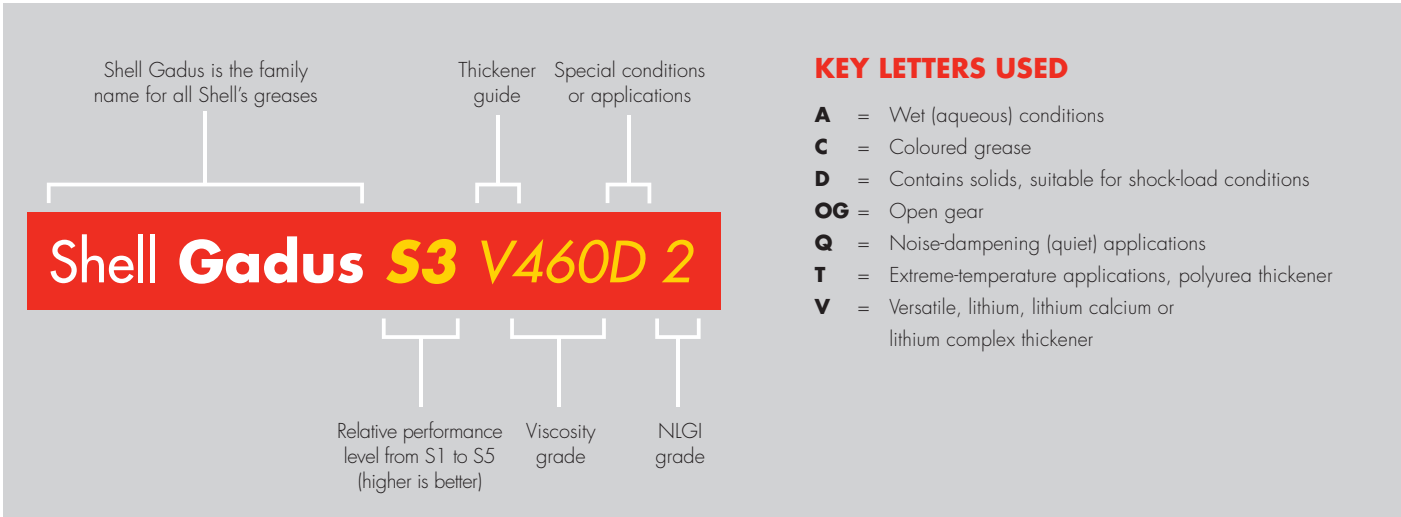


HELPFUL PRODUCT NAMES

There are at least four properties that you should look for when selecting a grease:

- thickener type
- oil viscosity
- operating conditions and application
- NLGI grade (consistency).

Shell's Gadus greases have names based on these properties that aim to help you select the right products for your needs.



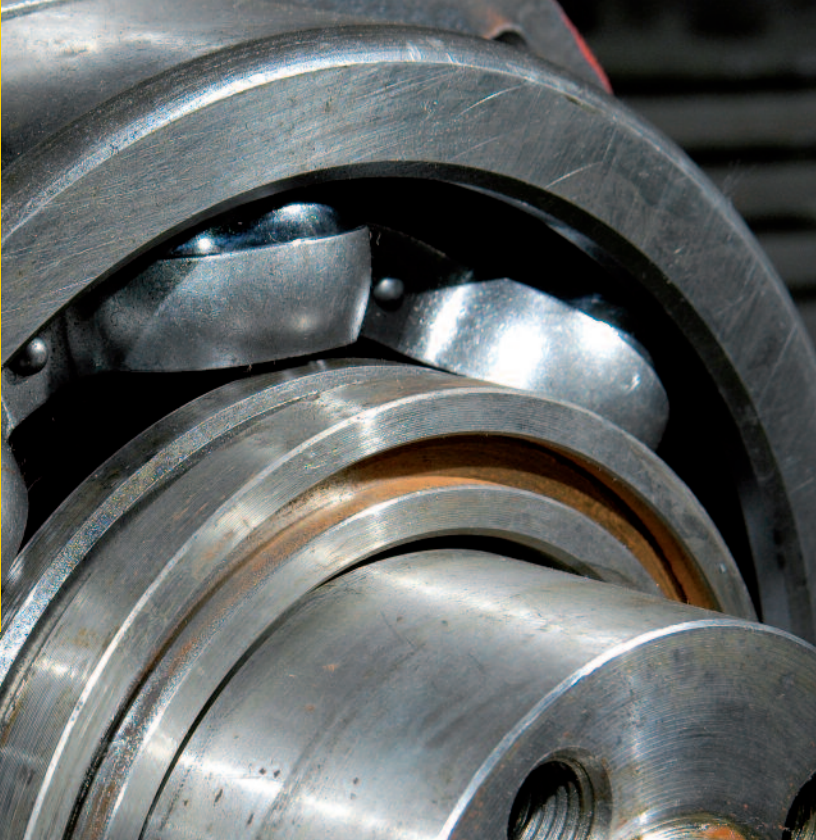
FULL PRODUCT AND SERVICE PORTFOLIO

Shell Lubricants is the number-one finished lubricants supplier² and has a 60-year history of innovation. It constantly invests to develop better lubrication solutions, including advanced synthetic technologies such as

- Shell Tellus S4 ME synthetic hydraulic oil – for long life and energy saving
- Shell Corena S4 R air compressor oil – for up to 12,000 hours of protection.

In addition, Shell provides the world-leading Shell LubeAnalyst oil condition monitoring service, which is designed to help improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.



Product	Benefits	Technology (Thickener, base oil and key properties)	ISO viscosity grade	NLGI grade	Specifications and approvals (Ask your Shell representative for full approval details. Approvals and claims vary by viscosity grade.)
Shell Gadus “V” range					
Shell Gadus S5 V100	■ Long life ■ Improved efficiency ■ Lithium complex	Lithium complex, synthetic	100	2	Approved: Vestas VWind Systems Listings: ABB, CH Schäfer Getriebe, FLSmidth, HegenscheidtMFD and perma Schmiersysteme
Shell Gadus S3 V220C	■ Extra protection ■ High temperature ■ Red lithium complex	Lithium complex, mineral, EP	220	1, 2	Specification: Meets ASTM D4950-08 GC1B. Approved: Aichelin, Dexter Axle, Ser-Norge Aluminium and Willy Vogel Listings: Listed by over 20 major companies from a wide variety of industries.
Shell Gadus S3 V460D	■ Heavy-duty protection ■ High temperature ■ Lithium complex	Lithium complex, mineral, EP, solid EP additive (MoS ₂)	460	2	Specification: Meets Caterpillar, Liebherr and P&H Mining Equipment Approved: Bucyrus International, CMI FPE Ltd, Gebr. Pfeiffer perma Schmiersysteme, Rothe Erde, SMS Meer and Voith Listings: BE (certified), Bucyrus International, CMI Groupe, Dieffenbacher, Gebr. Pfeiffer, Hitachi, Konecranes and Komatsu Mining
Shell Gadus S2 V100	■ Reliable protection ■ Multipurpose ■ Lithium	Lithium, mineral	100	1, 2, 3	Approved: Approved by 18 major companies from a wide variety of industries. Listings: Listed by over 60 major companies from a wide variety of industries.
Shell Gadus S2 V220	■ Reliable protection ■ Multipurpose ■ Lithium	Lithium, mineral, EP	220	00, 0, 1, 2, 3	Specification: Meets ASTM D4950-08 LB Approved: Approved by 24 major companies from a wide variety of industries. Listings: Listed by over 100 companies from a wide variety of industries.
Shell Gadus S2 V220AD	■ Heavy-duty protection ■ Water resistant ■ Lithium calcium	Lithium calcium, mineral, EP, solid EP additive (MoS ₂)	220	1, 2	Approved: Sandvik (endorsed) and Xiamen XGMA Machinery Listings: Listed by 12 major companies from a wide variety of industries.
Shell Gadus S2 V220AC	■ Heavy-duty protection ■ Water resistant ■ Red lithium calcium	Lithium calcium, mineral, soluble EP additive	220	0, 1, 1.5, 2, 3	Approved: Willy Vogel Specification: Meets ASTM D4950-08 LB Listings: David Brown, Danieli, Eirich, Harburg-Freudenberger Maschinenbau, Maschinenfabrik Gustav Eirich, New Zealand Steel, Ondrives, Power Jacks Group, Tetra Pak International, ZZ-Antriebe
Shell Gadus speciality ranges Contact your Shell representative for details					
Shell Gadus “Q” greases for noise-dampening (quiet) applications					
Shell Gadus automotive greases for automotive components, from door-lock motors to steering systems and constant-velocity joints					
Shell Gadus “OG” lubricants for mobile and stationary open-gear applications requiring spray or fluid bath lubrication					
Shell GadusRail greases meet European and American railway specifications, especially for railway applications, including axle journal bearings, traction motor bearings and curve rails					
Shell Tactic EMV automatic single-point lubricators – available with a selected range of high-performance Gadus greases					
Related products Contact your Shell representative for details					

FOR MORE INFORMATION, PLEASE CONTACT

“Shell Lubricants” refers to the various Shell companies engaged in the lubricants business.

¹Saving reported by one customer. Actual savings may vary, depending on the application, the current oil used, the maintenance procedures and the condition of the equipment.

²Source: Kline & Company, “Competitive Intelligence for the Global Lubricants Industry, 2008–2018.”



shell.com/lubricants



SHELL GADUS IS A COMPREHENSIVE FAMILY OF GREASES DESIGNED TO MEET YOUR NEEDS.

Shell Gadus is a comprehensive family of greases designed to meet your needs. The Gadus ranges include multipurpose greases that are designed to help simplify your product inventory, as well as speciality greases, including advanced polyurea synthetic products designed for the most severe extreme-temperature and long-life applications, and a range of open-gear lubricants.

Whether you need greases for steel production, mining, construction, power generation, general or automotive manufacturing, other applications or your vehicles, Shell has a grease designed to best meet your challenges.

The Shell Gadus range has been developed to deliver optimum value through

- enhanced wear protection
- long grease life
- system efficiency.

WEAR PROTECTION

Equipment wear can reduce system efficiency and service life. Protecting components from wear is fundamental for getting the most out of your investments through prolonging asset life and preventing production losses through breakdowns.

The Shell Gadus range of greases offers protection across a wide range of applications. The range includes Shell's latest extreme-temperature "T" polyurea synthetic greases, which can help to protect bearings under extreme temperatures, heavy loads and contaminated conditions.

LEADING TECHNOLOGY

Shell's international research and development team works in close co-operation with customers and equipment makers to create the latest advanced-technology greases. These greases are designed to outperform conventional greases by combining a wide range of thickener technologies, which keep the grease in place under different conditions, with innovative oils that provide efficient lubrication and protection.



GREASE LIFE

The longer your grease lasts, the longer your components last and the less lubrication maintenance your equipment needs. Shell Gadus greases are designed to help your equipment continue to operate without interruption. The comprehensive product range enables you to select a grease with a life that meets your operational needs.

Customers using Shell Gadus extra-long-life synthetic technologies are doubling grease life for some applications. Others prefer to take advantage of cost-effective, reliable, multipurpose Shell Gadus greases for standard life applications.

SYSTEM EFFICIENCY

To help your equipment perform to its design standards, you need a grease that stays in place and provides effective protection and lubrication where you need it.

Shell Gadus greases are designed to help maintain or even improve the efficiency of your systems. From greases that offer reliable performance for standard applications to specialist low-noise, high-temperature and heavy-load greases, there is a Shell Gadus product that can help to optimise your system's efficiency and costs of operation.

A WIDE RANGE OF THICKENER TECHNOLOGIES

Different applications and conditions require different thickeners. The Shell Gadus range uses a wide range of thickener technologies, each with relative benefits.

Thickener guide	Thickener	Temperature	Water resistance	Shear and vibration stability
T	Polyurea	✓✓✓✓✓	✓✓✓	✓✓✓✓✓
V	Lithium complex	✓✓✓✓	✓✓✓	✓✓✓✓✓
V	Lithium calcium	✓✓	✓✓✓✓	✓✓✓
V	Lithium	✓✓	✓✓	✓✓✓
A	Calcium	✓	✓✓✓	✓✓✓
U	PTFE	✓✓✓✓✓	✓✓✓✓✓	✓✓✓✓
U	Clay	✓✓✓✓✓	✓✓	✓✓✓

- T** = Extreme-temperature performance using Shell polyurea thickeners
V = Versatile, multipurpose application using lithium and lithium complex thickeners
A = Thickeners for wet (aqueous) conditions
U = Unusual non-melting thickeners or other applications

SHELL'S ADVANCED POLYUREA "T" THICKENER

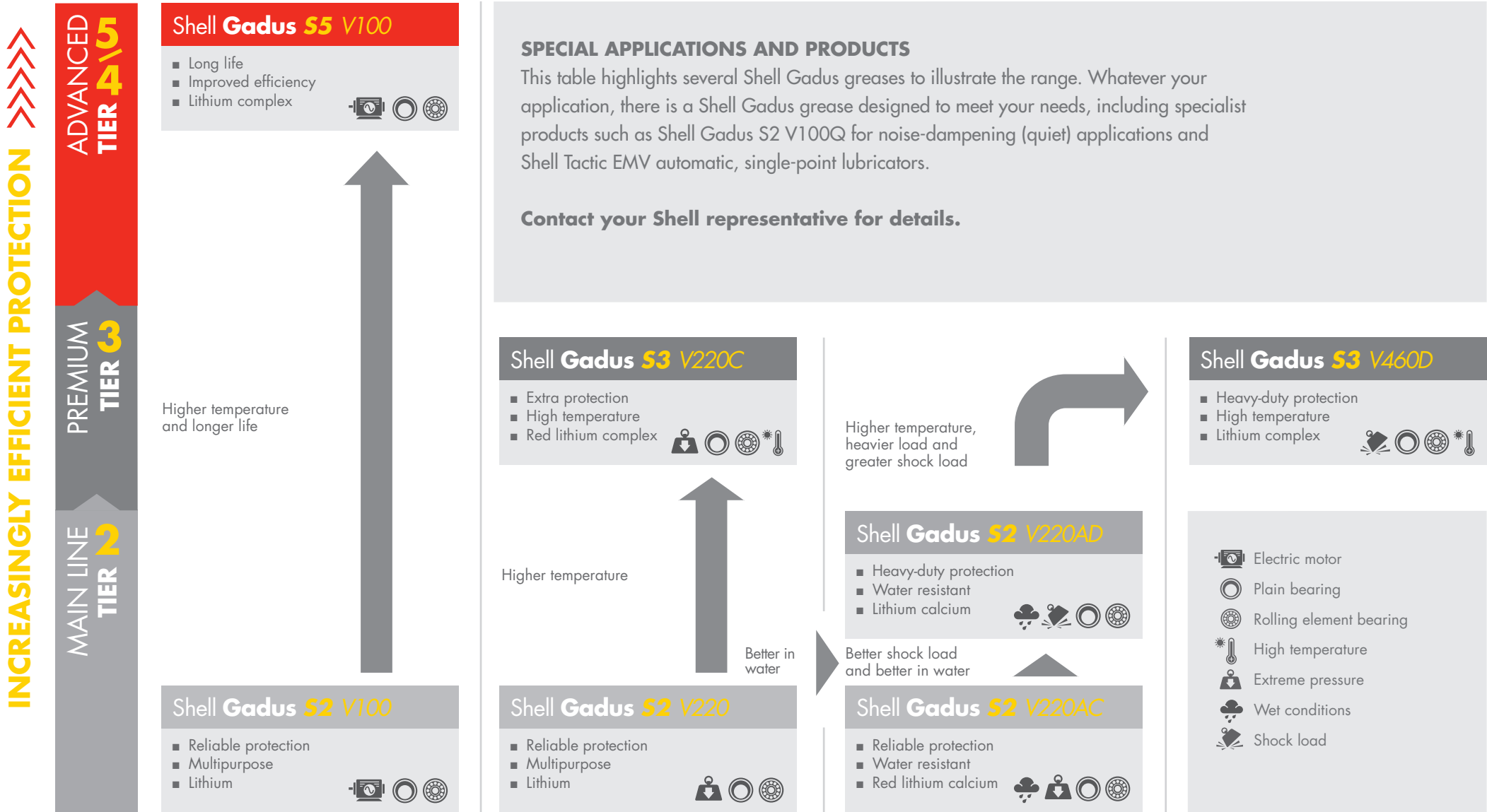
Shell's research teams have developed superior greases using advanced polyurea thickener technology. These products are designed to outperform conventional greases in extreme-temperature applications, such as electric motors and automotive constant-velocity joints. In one example, a global steel manufacturer reported that it reduced bearing failure rates by 75% and grease consumption by 30% through switching to Shell Gadus S3 T100 polyurea technology grease.

FIT-FOR-PURPOSE PROTECTION

Using the right grease for your needs can deliver significant savings. For example, a construction company reports saving \$3.8 million¹ a year through changing to Shell Gadus S3 V460D 2 grease, which provided the protection needed for performing heavy-duty work. The grease is extending excavator component life by 40%, reducing downtime and maintenance costs, and increasing equipment availability.

A RANGE OF GREASES TO MEET YOUR NEEDS

MULTIPURPOSE Bearings carrying low loads at higher speeds Auxiliary industrial equipment, including electrical motors and fans, and lightly loaded roller element or plain bearings	MULTIPURPOSE EXTREME PRESSURE (EP) Bearings carrying medium loads at medium speeds Roller element or plain bearings for industry and transport		HEAVY-DUTY MULTIPURPOSE Bearings carrying heavy loads at lower speeds and shock loads Large roller element or plain bearings for industry and transport
	DRY ENVIRONMENTS	WET ENVIRONMENTS	
	GREASE BASE OIL VISCOSITY 100	220	



SHELL TACTIC EMV LUBRICATORS

Irregular greasing can cause problems. Using a Shell Tactic EMV automatic, single-point lubricator, especially in hard-to-reach places, can help to enhance operational efficiency through precise and reliable lubrication. For example, a mining customer reports saving over \$167,000¹ a year through using the Shell Tactic EMV system with a high-quality Shell Gadus grease to increase maintenance efficiency.

Product	Benefits	Technology	ISO viscosity grades	Specifications and approvals (Full details of approvals for all products can be obtained from your Shell representative; approvals and claims will vary by viscosity grade.)
Shell Omala S4 GX	■ Extra protection ■ Extra life ■ Special applications	Synthetic (advanced extreme-pressure, EP, system)	150, 220, 320, 460	Approved by or meets David Brown, GE (wind turbines), Gamesa and many other equipment manufacturers Industry standards: ANSI/AGMA 9005-E02 (EP); ISO 12925-1 CKD; DIN 51517-3 (CLP); US Steel 224
Shell Omala S4 VVE	■ Extra protection and life ■ Energy saving ■ Worm drives	Synthetic (polyalkylene glycol)	150, 220, 320, 460	Approved by or meets Bonfiglioli, David Brown and many other equipment manufacturers Industry standards: ANSI/AGMA 9005-E02 (EP); ISO 12925-1 CKE
Shell Omala S2 G	■ Extra protection ■ Standard applications	Conventional (EP)	68, 100, 150, 220, 320, 460, 680, 1000	Approved by or meets David Brown, Cincinnati Machine, and many other equipment manufacturers Industry standards: ANSI/AGMA 9005-E02 (EP); ISO 12925-1 CKD; DIN 51517-3 (CLP); US Steel 224
Shell Omala S1 VV	■ Reliable protection ■ Worm drives	Conventional (compounded fatty oil)	460	Industry standards: AGMA 9005-E02 (CP)
Speciality gear oils				
Shell Spirax oils	A range of automotive gear, axle and transmission oils to meet the needs of on- and off-highway vehicles			Contact your Shell representative for details

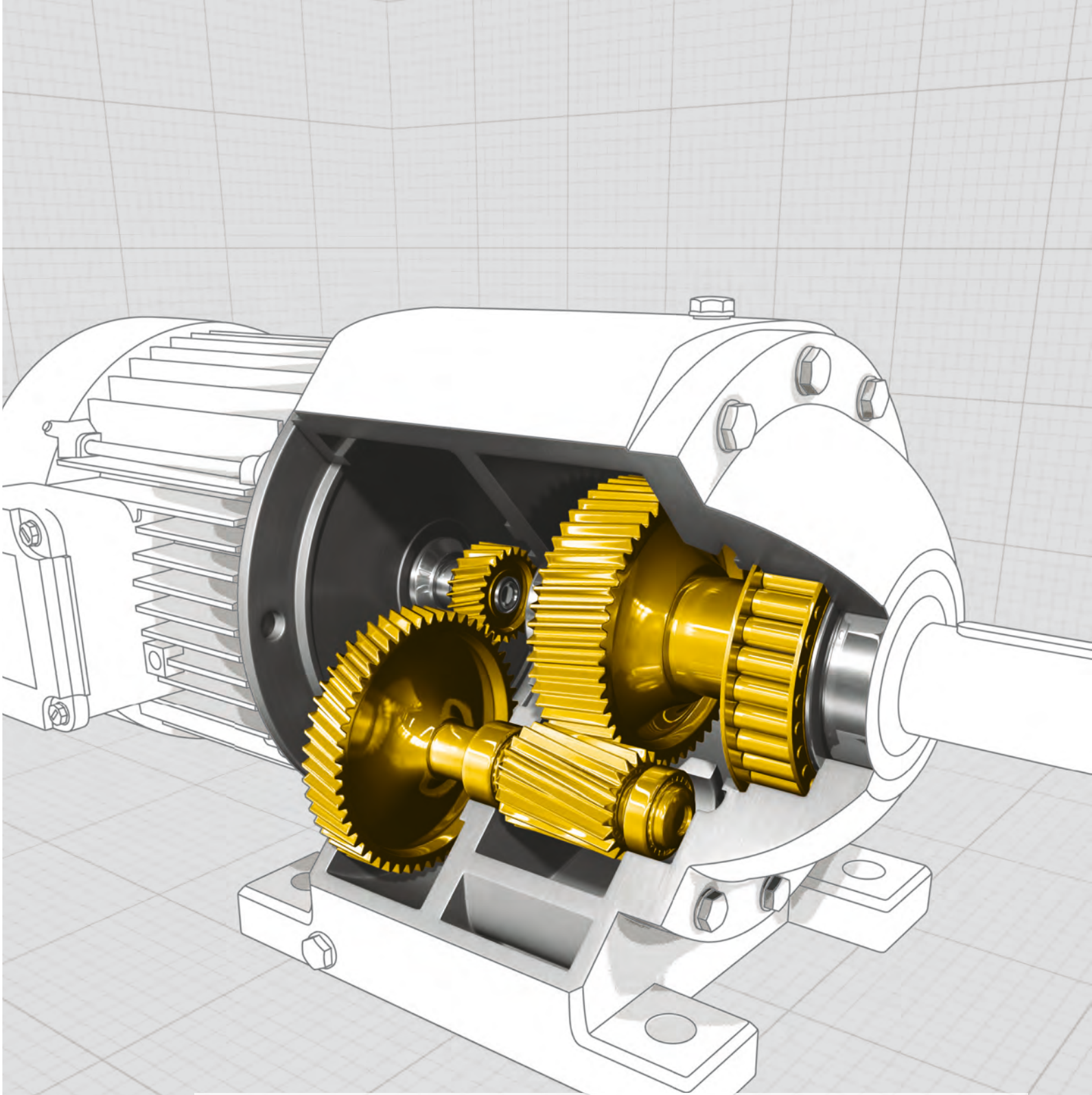
FULL PRODUCT AND SERVICE PORTFOLIO

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- Shell Tellus S4 ME synthetic hydraulic oil – for long life and energy saving
- Shell Corena S4 R air compressor oil – for up to 12,000 hours of protection.

In addition, Shell provides the excellent Shell LubeAnalyst oil condition monitoring service, which is designed to help improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.



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²Source: Kline & Company, "Competitive Intelligence for the Global Lubricants Industry, 2008–2018.

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The Shell Omala range of industrial gear oils has been developed to enable equipment operators to select the oil that will deliver optimum value to their operations through enhanced wear protection, long oil life and high system efficiency.

WEAR PROTECTION

Protecting machine gears from wear and corrosion is important for prolonging asset life and preventing breakdowns. The Shell Omala range of industrial gear oils offers protection across a wide range of applications: from the extreme-pressure rolling contacts in spur gears to the sliding contacts in worm gears. It includes Shell's latest synthetic oils such as Shell Omala S4 GX, which offers exceptional long-life performance under extreme loads and temperatures.

OIL LIFE

The longer an oil's life, the less fluid maintenance your equipment requires, so it can continue to operate without interruption. The Shell Omala range enables you to match oil life to your operational needs. Customers using Shell's standard products, such as Shell Omala S2 G, are extending oil-drain intervals by up to 200% for some applications. Customers are trading up to synthetic oils, such as Shell Omala S4 GX, for even longer oil life and other benefits.

SYSTEM EFFICIENCY

Shell Omala oils can help to maintain or improve system efficiency by protecting against the impact of oil ageing and contamination for efficient lubrication. Advanced synthetic oils such as Shell Omala S4 WE can offer additional benefits by providing improved energy efficiency compared with conventional oils.

A RANGE OF INDUSTRIAL GEAR OILS DESIGNED TO MEET YOUR NEEDS

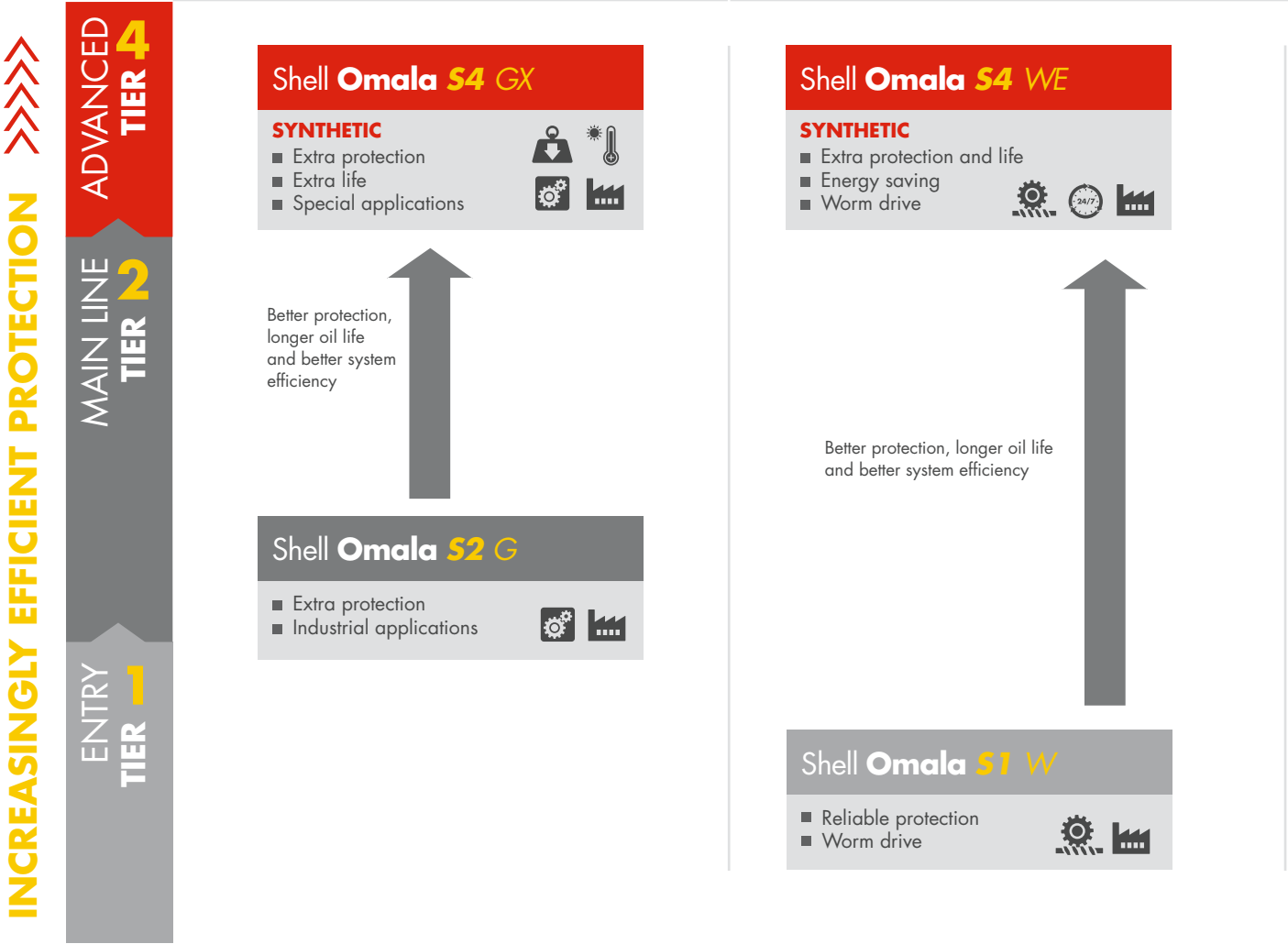
To meet the challenges of a wide range of gear types and applications, Shell has designed a portfolio of oils that enables you to choose a product to best match your technical and operational needs.

PRODUCT NAME SUFFIX KEY

- E = Energy saving, high efficiency
- G = Gears – spur and helical
- W = Worm drive
- X = Extra/extreme performance

APPLICATION ICON KEY

- Factory/machine applications
- Enclosed gears
- High temperature
- Extreme load
- Long life
- Worm gear



PERFORMANCE YOU CAN COUNT ON

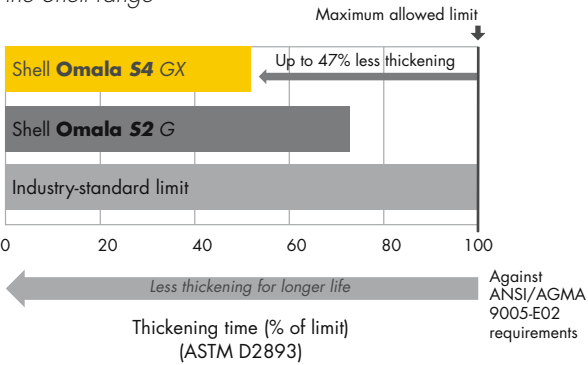
Shell's industrial gear oils are developed in close co-operation with customers and equipment makers. The Shell Omala range is approved or listed by hundreds of equipment makers, and its performance has been demonstrated in real-life applications to help reduce the possibility of your equipment letting you down.

REAL-WORLD VALUE DELIVERY

Users of Shell Omala oils across a wide range of industries are benefiting from proven, robust and reliable lubrication that adds value to their operations. For instance, by switching to Shell Omala S4 GX, some customers are

- increasing equipment protection: one customer reports virtually eliminating gearbox failures and saving over \$50,000 a year
- extending oil-drain intervals. Some equipment manufacturers approve Shell Omala S4 GX for four-year drain intervals, and some customers have extended their oil-drain intervals by up to five times.
- enhancing operational efficiency. Wind-turbine operators have increased turbine availability by reducing shutdowns due to foaming, high temperatures and filter blocking.

Shell Omala S4 GX – For the longest oil life in the Shell range



Product	Benefits	Specifications and approvals (Full details of approvals for all products can be obtained from your Shell representative.)
Gearboxes and manual transmissions		
Shell Spirax S6 GXME	■ Exceptional performance ■ Energy efficient ■ Maintenance saving	Exceeds requirements of API GL-4 and MT-1; meets requirements of ZF TE-ML-02D Approved by MAN 341 Type E4; and Volvo 97307
Shell Spirax S2 G	■ Proven performance	Meets requirements of API GL-4
Axles		
Shell Spirax S6 AXME	■ Exceptional performance ■ Energy efficient ■ Maintenance saving	Meets requirements of API GL-5 and MT-1; MIL-PRF-2105E; and SAE J2360 Approved by AGMA Standard 250.03; ArvinMeritor O76-N, O76-E, O76-M, O76-Q and O76-R; DAF; Dana SAES-256 Rev. C; GE D50E9C; International TMS 6816; Mack GLJ and GOJ Plus; MAN 342 Type S1; Scania STO 1.0; US Steel 224; Volvo 97312; ZF TE-ML 05B, 07A, 12B, 16F, 17B, 19C and 21B
Shell Spirax S3 AX	■ Proven performance ■ Key OEM specifications	Exceeds requirements of API GL-5 and MT-1; meets requirements of MIL-2105D; MAN 342 Type M2; and ZF-TE-ML
Shell Spirax S2 A	■ Proven performance	Meets requirements of API GL-5
Automatic transmissions		
Shell Spirax S4 ATF HDX	■ Proven performance ■ Versatile application	Meets requirements of Ford Mercon®; General Motors Dexron® III G; Allison C4; Mercedes-Benz 236.9; 339 Type Z2 and 339 Type V2; Voith 55.6336 (ex, G1363); and ZF TE-ML 04D, 09, 14B, 16L and 17C
Shell Spirax S2 ATF D2	■ Proven performance	Meets requirements of General Motors Dexron® II and Allison C4

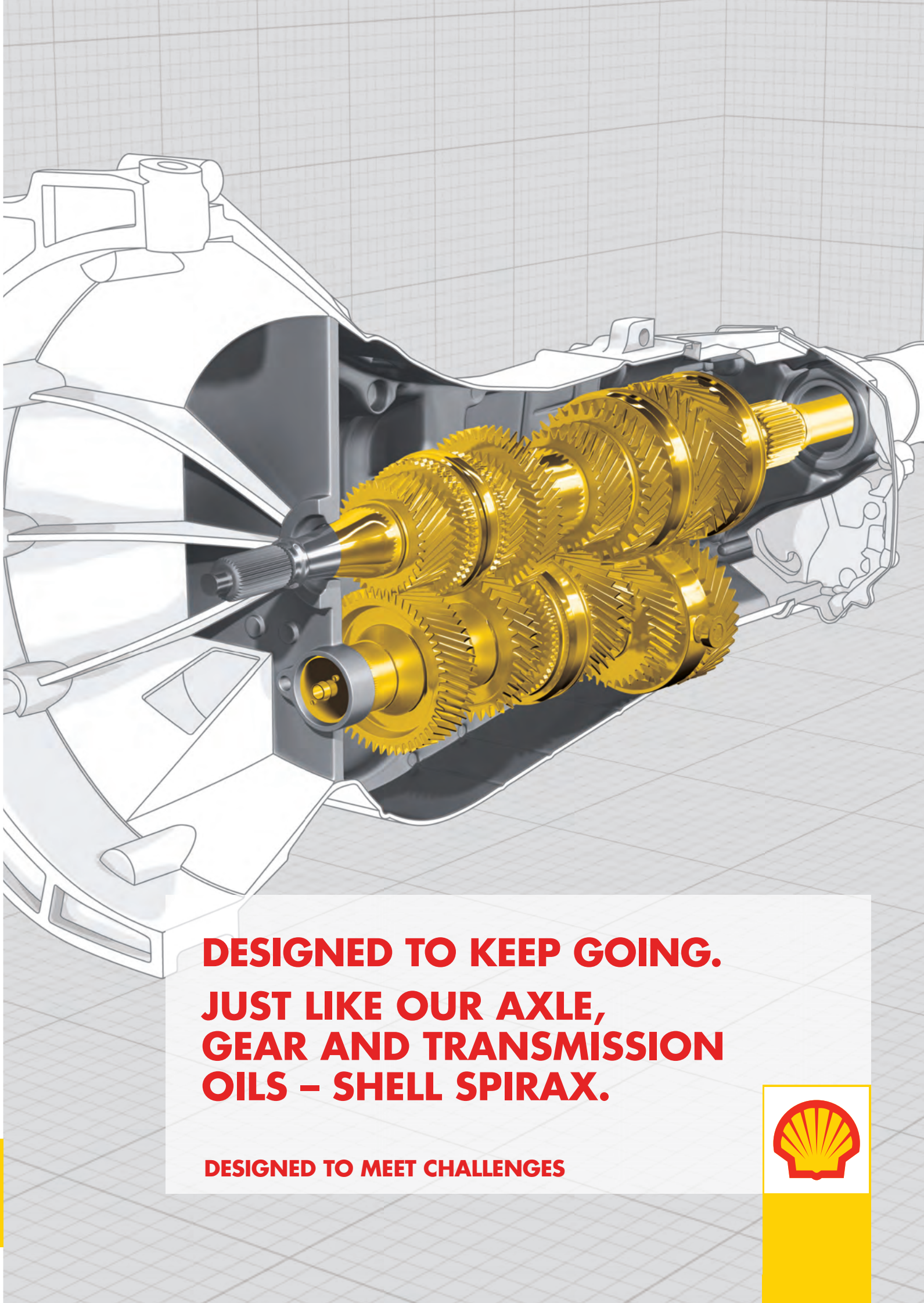
FULL PRODUCT AND SERVICE PORTFOLIO

Shell Lubricants is named the number one lubricants supplier [Kline & Company, 2009] and has a 60-year history of innovation. It constantly invests to develop better lubrication solutions, including advanced synthetic technologies such as

- Shell Rimula R5 LME synthetic heavy-duty engine oil
- Shell Spirax S6 AXME synthetic axle oil.

In addition, Shell provides the excellent Shell LubeAnalyst oil condition monitoring service, which is designed to help improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.



DESIGNED TO KEEP GOING.
JUST LIKE OUR AXLE,
GEAR AND TRANSMISSION
OILS – SHELL SPIRAX.

DESIGNED TO MEET CHALLENGES



For more information, please contact

shell.com/lubricants

“Shell Lubricants” refers to the various Shell companies engaged in the lubricants business.

EVERY PART OF YOUR VEHICLE HAS BEEN METICULOUSLY ENGINEERED, SO YOU WANT TO BE SURE THAT YOU CHOOSE A LUBRICANT THAT HAS BEEN DESIGNED TO ENSURE YOUR EQUIPMENT IS WELL PROTECTED AND WORKS EFFICIENTLY.

The Shell Spirax range of axle, gear and automatic transmission fluids has been developed to enable drivers, fleet owners and owner operators to select the oil that will deliver optimum value to their operations through enhanced wear protection, long oil life and efficiency.

PROVEN WEAR PROTECTION

Protecting axles and gears from wear can help to reduce maintenance costs, extend vehicle life and maximise your return on investment.

The Shell Spirax range of axle and gear oils offers proven protection, including the latest synthetic oils that deliver exceptional wear, pitting, bearing-failure and corrosion protection for heavy-duty axles, transmissions and gears operating under highly stressed conditions. This can help to reduce your costs.

LONGER OIL AND COMPONENT LIFE

The longer an oil goes on protecting its components, the less maintenance your vehicle requires, so it can continue to operate with less interruption, which can reduce your maintenance costs. The Shell Spirax range contains oils that are designed to offer you value through exceptional oil life and protection that enable oil-drain intervals and component life to be extended. For example, in industry-standard oxidation tests, Shell Spirax S6 AXME axle oil remains in specification after four times the normal duration of the test, which means that it can go on protecting your vehicle components for longer.

EFFICIENCY

Shell Spirax oils are designed to help keep your transmission components protected so that they can go on working efficiently. For instance, Shell Spirax S6 AXME and Shell Spirax S6 GXME have special frictional properties and high fluidity that reduce power loss, lower the operating temperature and offer higher mechanical efficiency. These properties can help to deliver fuel efficiency and savings.

PERFORMANCE YOU CAN COUNT ON

The benefits of Shell Spirax S6 GXME and Shell Spirax S6 AXME have been demonstrated through extensive laboratory tests and over 20 million kilometres of field trials. Shell's axle and gear oils are developed in close co-operation with customers and equipment manufacturers. The Shell Spirax range is approved by many vehicle and automotive component manufacturers.

PRODUCT-NAME SUFFIX KEY

- A

=

Axle
- ATF

=

Automatic transmission fluid
- D

=

Daimler/Dexron®
- E

=

Energy efficiency
- G

=

Gear
- HD

=

Heavy duty
- LS

=

Limited slip
- M

=

(Low) maintenance
- M

=

Mercon®/MAN
- U

=

Allison
- V

=

Voith
- X

=

Multi-vehicle/multi-OEM
- Z

=

ZF

APPLICATION ICON KEY

- Heavy truck
- Tractor
- Motorbike
- SUV
- Pickup
- Truck
- Heavy truck cabover
- Passenger car
- Excavator
- Bus/coach

INCREASINGLY EFFICIENT PROTECTION

ADVANCED TIER 5/6
PREMIUM TIER 3/4
MAIN LINE TIER 2

GEARBOXES AND MANUAL TRANSMISSIONS

Shell Spirax S6 GXME

SYNTHETIC

■ Exceptional performance

■ Energy efficient

■ Maintenance saving

Greater fuel efficiency and lower maintenance

Shell Spirax S2 G

■ Proven performance

AXLES

Shell Spirax S6 AXME

SYNTHETIC

■ Exceptional performance

■ Energy efficient

■ Maintenance saving

Greater fuel efficiency and lower maintenance

Shell Spirax S3 AX

■ Proven performance

■ Designed for OEMs/MAN/Daimler

Shell Spirax S2 A

■ Proven performance

AUTOMATIC TRANSMISSIONS

Shell Spirax S4 ATF HDX

■ Proven performance

■ Versatile application

Shell Spirax S2 ATF D2

■ Proven performance

■ Dexron® IID

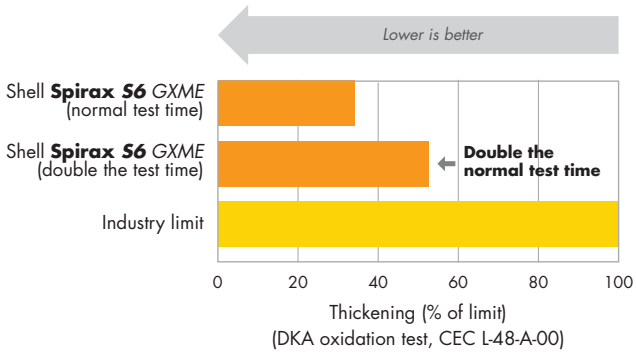
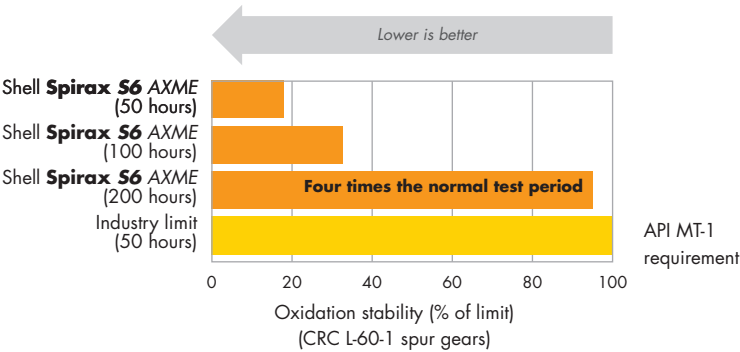
Contact your Shell representative to find out about the full range of Shell Spirax axle, gear and automatic transmission fluids, including products specifically designed for the construction and agriculture industries

THE BENEFITS OF USING DEDICATED AXLE AND GEAR OILS

Many gearboxes and axles use different materials and operate under different conditions. A single oil used for both applications may lead to a compromise in either or both the axle and gearbox wear protection and efficiency. Shell Spirax oils have been specifically designed to meet the challenges and requirements of many industry-leading automatic transmission, axle and gear manufacturers.

The oils are formulated by Shell's scientists to help deliver the specific oxidation, shear stability and performance characteristics needed to best provide protection, power and efficiency for each application.

Shell Spirax S6 GXME and Shell Spirax S6 AXME – For the longest oil life in the Shell range



Product	Benefits	Technology	ISO viscosity grades	Specifications and approvals (Full details of approvals for all products can be obtained from your Shell representative; approvals and claims will vary by viscosity grade.)
Shell Tellus S4 ME	■ Extra long life ■ Energy saving	Synthetic, ashless	HM/22, 32, 46, 68	Approved by Denison Hydraulics, Cincinnati Machine, Eaton (Vickers), Bosch Rexroth and many other equipment manufacturers Industry standards: ASTM D6158, ISO 11158, DIN 51524-2
Shell Tellus S3 M	■ Long life and improved protection ■ Industrial applications	Mineral, ashless	HM/22, 32, 46, 68, 100	Approved by Denison Hydraulics, Eaton (Vickers), Cincinnati Machine and many other equipment manufacturers Industry standards: ISO 11158, DIN 51524-3, ASTM D6158
Shell Tellus S2 V	■ Extra protection ■ Versatile application	Mineral, zinc-based	HV/15, 22, 32, 46, 68, 100	Approved by Denison Hydraulics, Cincinnati Machine, Eaton (Vickers) and many other equipment manufacturers Industry standards: ISO 11158, ASTM D6158-05, DIN 51524-3
Shell Tellus S2 M	■ Extra protection ■ Industrial applications	Mineral, zinc-based	HM/22, 32, 46, 68, 100	Approved by Denison Hydraulics, Cincinnati Machine, Eaton (Vickers), Bosch Rexroth and many other equipment manufacturers Industry standards: ISO 11158, ASTM D6158-05, DIN 51524-2
Shell Hydraulic S1 M	■ Reliable protection ■ Industrial applications	Mineral, zinc-based	HM/32, 46, 68	ISO 11158
Speciality grades				
Shell Tellus S4 VX	■ Ultra-low temperature ■ Versatile application	Special base oils, ashless	HV/32	Approved by Komatsu Mining, Komatsu and DIETZ automation (servo valve and proportional valve test equipment)
Shell Tellus S2 VA	■ Extra protection ■ Water tolerant	Mineral, zinc-based, detergent	LHV/46	Industry standards: DIN 51502, ISO 6743/4, ISO 11158, ASTM 6158-05
Shell IruS	A full range of fire-resistant and less flammable fluids for special applications			Contact your Shell representative for details
Shell Naturelle	A full range of environmentally considerate hydraulic oils			Contact your Shell representative for details

FULL PRODUCT AND SERVICE PORTFOLIO

Shell Lubricants is named a global leader in lubricants (Kline & Company 2009) and has a 60-year history of innovation. It is constantly investing to develop better lubrication solutions, including advanced synthetic technologies such as

- Shell Omala S4 GX synthetic gear oil – for long life in demanding environments
- Shell Corena S4 R air compressor oil – for up to 12,000 hours of protection.

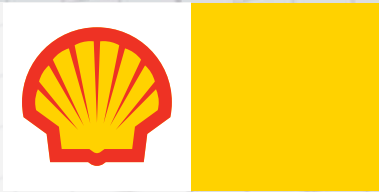
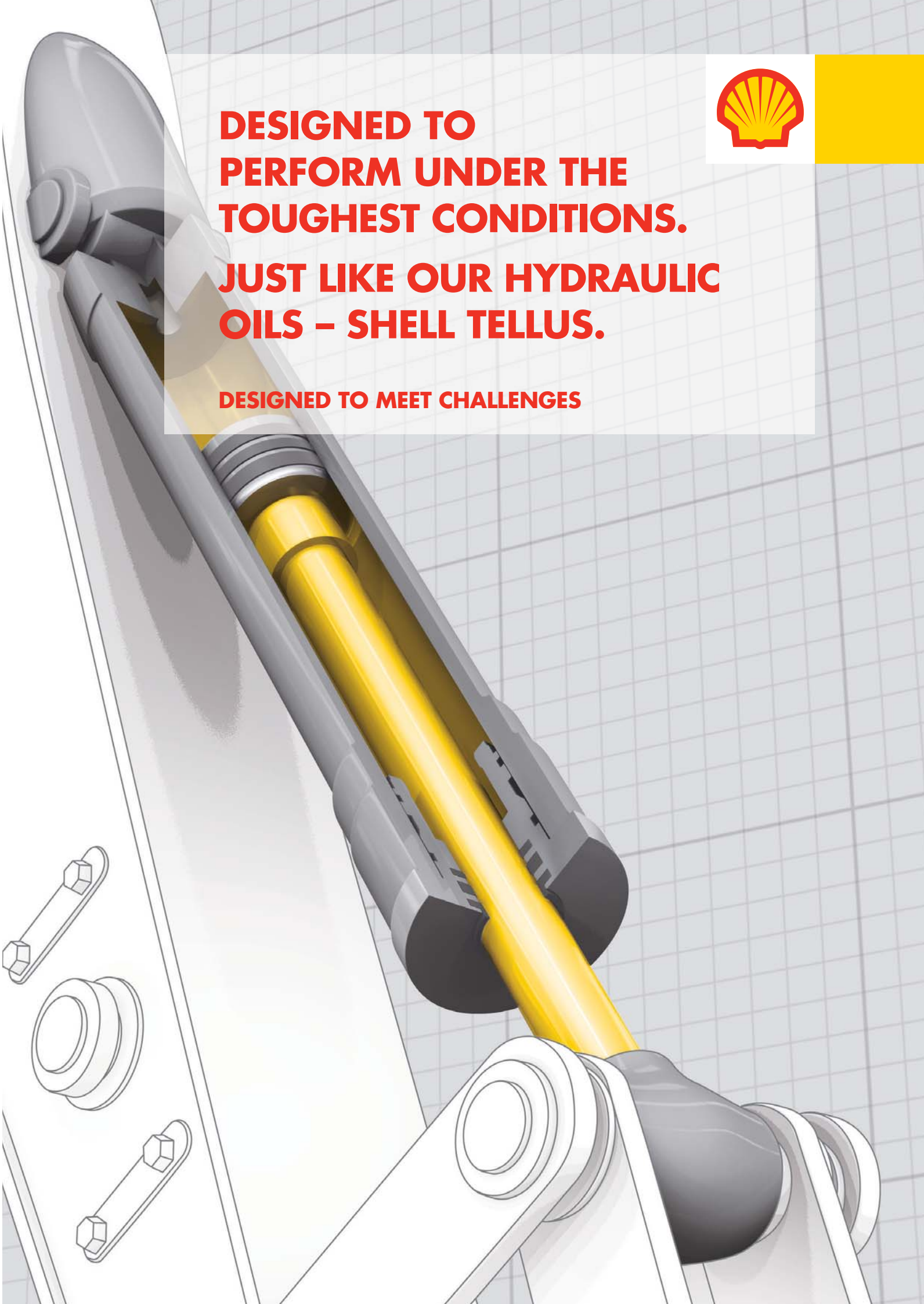
In addition, Shell provides the excellent Shell LubeAnalyst oil condition monitoring service, which is designed to help improve your business performance.

Whatever your needs or application, Shell can provide a full range of oils and greases, including synthetic, high-performance products and additional services.



For more information, please contact

shell.com/lubricants



DESIGNED TO
PERFORM UNDER THE
TOUGHEST CONDITIONS.
JUST LIKE OUR HYDRAULIC
OILS – SHELL TELLUS.

DESIGNED TO MEET CHALLENGES

EVERY PART OF YOUR MACHINE OR PROCESS HAS BEEN METICULOUSLY ENGINEERED, SO YOU WANT TO BE SURE THAT YOU CHOOSE A LUBRICANT THAT HAS BEEN DESIGNED TO ENSURE THAT YOUR EQUIPMENT IS WELL PROTECTED AND WORKS EFFICIENTLY.

The Shell Tellus range of hydraulic fluids has been developed to enable equipment operators to select the oil that can help deliver optimum value to their operations through enhanced wear protection, long oil life and high system efficiency.

WEAR PROTECTION

The hydraulic pump is the heart of your hydraulic system. Any wear can reduce efficiency and system service life. The Shell Tellus range of hydraulic fluids offers a range of options, including the latest synthetic, ashless technology that are designed to help extend pump life, even under the most severe conditions, through to proven, cost-effective, zinc-based additive technology for general use.

OIL LIFE

The longer the oil life, the less fluid maintenance is required to help your equipment operate for longer without interruption. The Shell Tellus range of hydraulic fluids enables you to match the oil life of the fluid to your operational needs. It includes extra-long-life synthetic technologies capable of up to four times standard life¹, through to cost-effective, reliable products for less-demanding applications.

SYSTEM EFFICIENCY

To help your equipment perform to its design standards, the hydraulic fluid needs to protect, lubricate and help transmit power in the most effective way possible. Shell Tellus hydraulic fluids are designed to help maintain or even improve the efficiency of hydraulic systems. From Shell Tellus S4 ME, which can improve the energy efficiency of many hydraulic systems, to Shell’s high-quality mainstream products that provide reliable air release, filterability and cleanliness, there is a choice that can help to optimise your system’s efficiency and costs of operation.

A RANGE OF HYDRAULIC FLUIDS DESIGNED TO MEET YOUR NEEDS

To meet the challenges of a wide range of hydraulic equipment and applications, Shell has designed a portfolio of fluids that enables you to choose a product to best match your needs.

PRODUCT-NAME SUFFIX KEY

- A** = Water tolerant
- E** = Energy saving, high efficiency
- M** = Manufacturing/machine: factory applications
- V** = Versatile applications
- X** = Extra/extreme performance

APPLICATION ICON KEY

- Factory/machine applications
- Long life
- High load
- Ultra-low temperature
- Mobile equipment/exterior use
- Wet conditions

INCREASINGLY EFFICIENT PROTECTION



THE SHELL TELLUS “M” RANGE

Industrial, manufacturing and machine use

Shell Tellus **S4 ME**

SYNTHETIC

- Extra long life
- Energy saving

Longer oil life, lower wear and better energy efficiency

Shell Tellus **S3 M**

- Long life and improved protection
- Industrial applications

Longer oil life and better protection

Shell Tellus **S2 M**

- Extra protection
- Industrial applications

Better wear protection and longer oil life

Shell Hydraulic **S1 M**

- Reliable protection
- Industrial applications

THE SHELL TELLUS “V” RANGE

Extended temperature operation for manufacturing and mobile machine use

Shell Tellus **S4 VX**

- Ultra-low temperature
- Versatile applications

Lower wear and better efficiency

Shell Tellus **S2 V**

- Extra protection
- Versatile applications

THE SHELL TELLUS SPECIALITY RANGE

Detergent-based fluids for use where water contamination is possible

Shell Tellus **S2 VA**

- Water tolerant
- Versatile applications

Better wear protection and system efficiency

REAL-WORLD VALUE DELIVERY

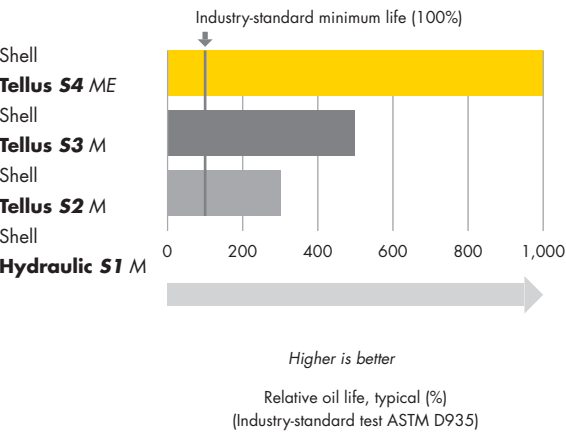
While many other companies design their fluids for the individual components of a hydraulic system such as the pump, Shell Lubricants looks at the system more broadly. For instance, Shell Tellus S4 ME has been designed to benefit the complete system. The result is a fluid that

- is statistically proven to help improve the energy efficiency of the system in which it is used. In a carefully controlled test cycle, one injection moulding equipment maker measured energy savings of up to 6.4%² in the hydraulic systems in which Shell Tellus S4 ME was used.
- can help to increase oil life by up to four times compared with conventional oils in the Shell range
- can help to prolong equipment life through excellent hydraulic pump wear protection.

IMPROVED CLEANLINESS

All Shell Tellus products now meet the demanding cleanliness requirements of the DIN industry standard to help provide extra protection, improved filter performance and less maintenance.

Shell Tellus S4 ME – For the longest oil life in the Shell range



¹ Compared with other oils in the Shell range
² The actual savings depend on the application, the current oil, the maintenance procedures, the equipment condition, the operating conditions and the intensity of hydraulic power usage.